

KARYASETU AI (कार्यसेतु)

Zero-Trust Gen AI Platform for Governed Content Transformation, Dual-Route Orchestration & Decentralized Provenance

One Trusted Source Document → Seven Governed, Audience-Tailored, Cryptographically Signed Deliverables in <25 Seconds

⚡ <25s Synthesis

🛡️ 0% Cloud Leakage

🔗 Ethereum Sepolia On-Chain

🔒 Ed25519 Sealed

💰 < ₹0.25 / Run

CRITICAL INDUSTRY CHALLENGES

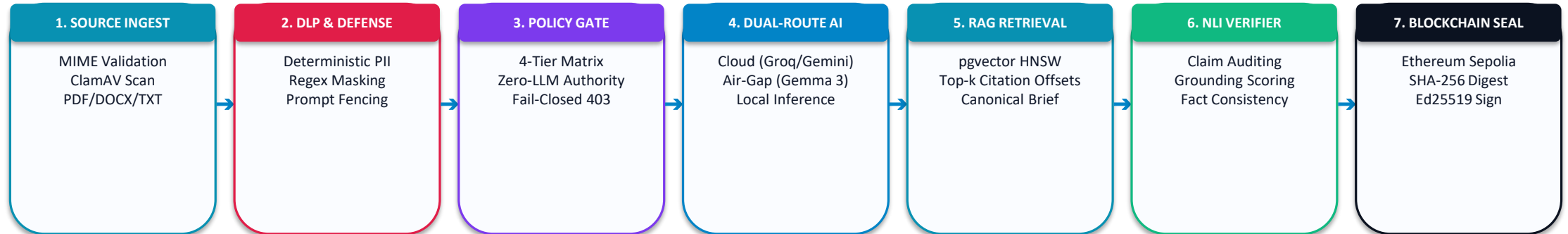
- ❌ **Sensitive Data Exposure:** Confidential enterprise & defense memos cannot be sent to public cloud LLMs (OpenAI/Claude) without violating data sovereignty and compliance laws.
- ❌ **Adversarial Prompt Injections:** Untrusted user documents and embedded indirect prompt instructions hijack downstream transformation pipelines and leak secrets.
- ❌ **Ungrounded Hallucinations:** Generative AI invents fictitious metrics and unverified claims when unanchored to exact source citation offsets.
- ❌ **Inter-Format Cross-Output Drift:** Manual rewriting into summaries, slides, and advisories leads to severe factual contradictions and metric discrepancies across formats.
- ❌ **Severe Manual Labor Bottleneck:** Teams spend 4–6+ hours per report manually synthesizing, formatting, and vetting collateral for distinct audiences.

KARYASETU ZERO-TRUST INNOVATION

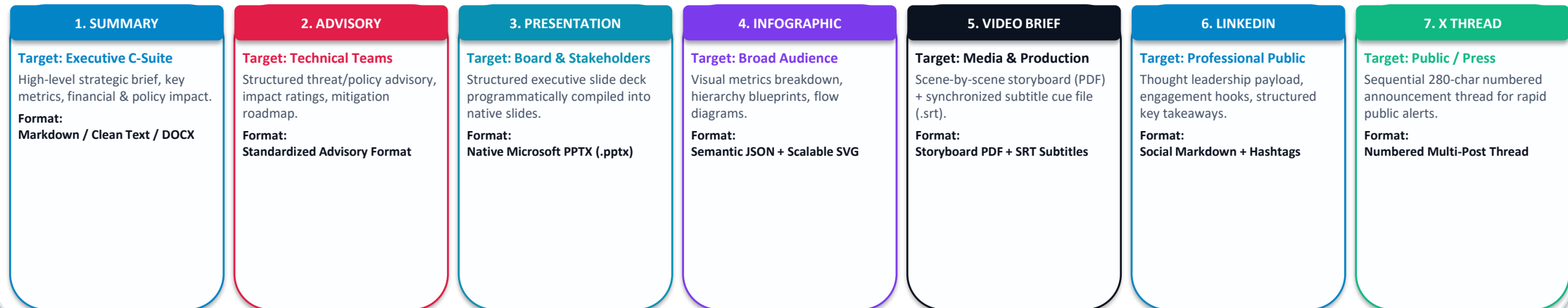
- ✅ **Dual-Route Air-Gapped AI:** Deterministic policy gateway enforces 100% offline local LLM execution (Gemma 3 12B) for Restricted/Confidential data; Groq/Gemini for Public.
- ✅ **Zero-Trust Ingress & DLP Defense:** Magic-byte verification, local ClamAV anti-malware scan, and deterministic regex PII redaction mask sensitive entities before AI tokenization.
- ✅ **Evidence-Grounded Vector RAG:** Dense semantic retrieval in pgvector binds every generated output claim to exact source chunk IDs with SHA-256 digests.
- ✅ **Single Canonical Fact Brief:** All 7 deliverables ground to one unified facts model, cross-verified by automated NLI entailment consistency checks to guarantee zero drift.
- ✅ **Decentralized Blockchain Provenance:** Every artifact is sealed with SHA-256 digests anchored directly to an Ethereum Sepolia smart contract and signed with Ed25519 keys.

PROPOSED SOLUTION: ZERO-TRUST PIPELINE & 7 GOVERNED DELIVERABLES

7-STAGE SECURITY & AI ORCHESTRATION PIPELINE



SEVEN GOVERNED, AUDIENCE-TAILORED DELIVERABLES



TECHNICAL APPROACH: 4-TIER ARCHITECTURE & CONCRETE STACK

TIER 1: USER EXPERIENCE & ENTERPRISE CONSOLE

Stack: Next.js 14 (App Router) • React 18 • TypeScript 5.7 • Tailwind CSS 3.4

Enterprise Trust Cockpit, Live Artifact Inspector, On-Demand Verification Tool, Real-time SSE streaming, and 38s Interactive Architecture Visualizer.

TIER 2: INGRESS DEFENSE, POLICY & ORCHESTRATION GATEWAY

Stack: FastAPI 0.115 Async • Python 3.12 • Pydantic 2.10 • LangGraph 0.2 • Redis 7 + Python-RQ

MIME magic-byte verification (%PDF-, PK), ClamAV daemon antivirus hooks, regex PII scrubbing (Aadhaar/PAN/Tokens), and deterministic 4-tier policy routing.

TIER 3: AI INFERENCE, RAG GROUNDING & NLI FACT VERIFICATION

Stack: Groq (openai/gpt-oss-120b) • Google Gemini API • Local Ollama/vLLM (Gemma 3 12B) • pgvector

Dense semantic retrieval over HNSW index, chunk citation anchors, prompt delimiter fencing, and NLI entailment scoring (SUPPORTED/CONTRADICTED/UNVERIFIED).

TIER 4: STORAGE, DECENTRALIZED PROVENANCE & LEDGER PLANE

Stack: PostgreSQL 16 + pgvector • MinIO / AWS S3 • Ethereum Sepolia Smart Contract • Ed25519 Web3.py

Immutable SHA-256 artifact digests anchored on Ethereum Sepolia contract (0x8AEf...7c08) with Ed25519 asymmetric signatures (RFC 8032) for non-repudiation.

CORE TECHNICAL & SECURITY SPECS

► **Deterministic Policy Routing:** Zero-LLM Authority: Policy engine is 100% pure Python. It decides whether content can use cloud or air-gapped local endpoints based on PUBLIC, INTERNAL, CONFIDENTIAL, RESTRICTED classification.

► **Air-Gapped Sovereign Route:** Sensitive data is physically restricted from cloud API egress; routed strictly to local Ollama / vLLM running Gemma 3 12B with 0% data leakage.

► **Fault Isolation Architecture:** Nested database savepoints isolate individual generator errors. If one deliverable encounters an issue, the remaining 6 continue cleanly.

► **EVM Sepolia Smart Contract:** Post-generation hook submits 32-byte content digests via Web3.py to Ethereum Sepolia. verifyDigest() enables public tamper detection.

► **Automated Fact Verification:** Natural Language Inference (NLI) cross-checks generated claims against source chunks, scoring grounding from 0.0 to 1.0.

FEASIBILITY, OPERATIONAL VIABILITY & RISK SAFEGUARDS

TECHNICAL FEASIBILITY

- **Asynchronous Queue Scaling:** Redis 7 + Python-RQ workers decouple generation from HTTP requests, eliminating timeouts on heavy PDF/PPTX builds.
- **Stateful Modular Isolation:** 7 generator adapters execute independently. Failure in one output never crashes remaining siblings.
- **Hybrid Model Interoperability:** Provider interface dynamically routes between Groq, Gemini, and local Ollama Gemma 3.

SECURITY & COMPLIANCE FEASIBILITY

- **NIST SP 800-207 Zero-Trust:** Pre-flight perimeter inspection: ClamAV anti-malware, magic-byte checks, and regex DLP PII scrubbing.
- **Air-Gapped Sovereign Enforcement:** Confidential institutional records physically forbidden from leaving private enclave networks.
- **Prompt Injection Fencing:** Strict instruction/data delimiters neutralize indirect prompt injection attacks.

OPERATIONAL & WORKFLOW FEASIBILITY

- **Dual-Key Human Approval Gate:** Mandatory human review checkpoint before external publication or social dissemination.
- **Destination-Aware Policies:** Restricted/internal outputs are hard-blocked from public social channels (LinkedIn/X).
- **SIEM-Ready Structured Audit:** JSON audit logs capture all authentication, policy decisions, and verification events.

ECONOMIC & FINANCIAL FEASIBILITY

- **Ultra-Low Cost per Transformation:** Groq / Gemini Flash costs < ₹0.25 per 7-deliverable run; local air-gapped runs cost ₹0.00 in API fees.
- **Zero External Vector DB Fees:** Native pgvector inside PostgreSQL 16 eliminates costly third-party Pinecone/Weaviate subscriptions.
- **Massive 98%+ Operational ROI:** Replaces 4–6+ hours of human specialist synthesis with <25s automated execution.

SECURITY & OPERATIONAL RISK MITIGATION MATRIX

Potential Risk Factor	System Vulnerability	KaryaSetu Concrete Mitigation Strategy
Prompt Injection & Hijacking	Embedded commands inside untrusted files	Strict delimiter encapsulation (<source_data>), tag neutralization, and input size clamping.
Factual Hallucination & Drift	Generators invent numbers/entities	pgvector chunk-level citation anchoring + automated NLI entailment cross-verification.
Confidential Data Exfiltration	Sensitive data egress to public cloud	Deterministic Python policy gate (zero LLM discretion); fails closed with HTTP 403 Forbidden.
Post-Generation Deliverable Tamper	Unauthorized document alteration	Immutable SHA-256 hash anchored to Ethereum Sepolia smart contract + Ed25519 signature.

INSTITUTIONAL IMPACT, COST STRUCTURE & VERIFIABLE PROOF

ECONOMIC VIABILITY & OPERATIONAL COST BREAKDOWN

Platform Component	Unit / Monthly Cost	Cost / 100 Runs	Operational Notes
Cloud LLM (Public Tier)	Groq/Gemini (~\$0.08/1M)	~\$0.60 – \$1.20	Generates all 7 deliverables per doc
Air-Gapped LLM (Confidential)	Local Gemma 3 12B	\$0.00 (Self-hosted)	Runs on existing enterprise GPUs
Blockchain Provenance	Sepolia Testnet / Polygon	<\$0.15 (~₹12 total)	45k gas per 32-byte digest anchor
Compute Infrastructure	Cloud VPS (2 vCPU, 4GB)	~\$15 – \$25 / month	Horizontally scalable RQ workers
Vector & Relational DB	PostgreSQL 16 + pgvector	~\$0 – \$25 / month	In-database HNSW cosine search
Object Storage (Artifacts)	MinIO / S3 on-premise	~\$1 – \$5 / month	Encrypted storage for PPTX/PDF/SVG
TOTAL RUN COST	< ₹0.25 / Document Run	~\$1.00 – \$2.00 total	Massive >98% cost savings vs manual

🔑 Business ROI Impact: Manual human synthesis of 500 documents/month consumes ~250 hours costing ₹2,50,000+. With KaryaSetu AI, the identical volume completes in under 5 hours total review time at ~₹3,500 total infrastructure cost (>95% operational ROI).

QUANTIFIABLE INSTITUTIONAL IMPACT

- ★ **10x Acceleration in Synthesis:** Transforms 50+ page technical briefs into 7 audience deliverables in under 25 seconds.
- ★ **Zero Data Leakage Guarantee:** Rigid deterministic policy enforcement prevents classified national intelligence from leaving sovereign infrastructure.
- ★ **Elimination of Multi-Channel Drift:** All deliverables ground to a single canonical fact brief with automated NLI consistency cross-checks.
- ★ **Complete Tamper Detection:** Ethereum Sepolia smart contract anchoring ensures post-generation deliverable integrity.

LIVE VERIFICATION LINKS & CITED STANDARDS

- 🔗 **Smart Contract (Sepolia Etherscan):** [0x8AEf680b6891E7e3cAdCBD4a499AbA1310F87c08](https://etherscan.io/address/0x8AEf680b6891E7e3cAdCBD4a499AbA1310F87c08)
- 🔗 **Relayer Signer Identity:** [0x460bb6AE2AD8a515d5E2886c69b78d8e3C2FBc41](https://etherscan.io/address/0x460bb6AE2AD8a515d5E2886c69b78d8e3C2FBc41)
- 🔗 **GitHub Repository:** <https://github.com/KetanGaikwadKRG/KaryaSetu>
- 🔗 **Official Evaluation Video:** <https://youtu.be/Lhi1ssHBfUk?si=T5blOI2TWDelf7DJ>
- 🔗 **Cited Standards & Literature:** NIST SP 800-207 (Zero Trust) • RFC 8032 (Ed25519) • W3C PROV-DM • IT Act Section 70